



THE EMBODIED SENSORY-MOTOR DATAFLOW & INTER-STAGE CONTRACTS

Part II: Transforming Raw Physical Transduction into Safe Physical Momentum across Five Cognitive Organs

THE PHYSICAL WORLD (W_t): CONTINUOUS ENVIRONMENTAL ENERGY FLUXES

Analog Photons (λ) · Contact Pressures (P) · Inertial Accelerations ($\mathbf{a}$) · Thermal Gradients (T) · Magnetic Fields ($\mathbf{B}$)

Physical Photons & Transduction

CHAPTER 4: PERCEPTION & SPATIAL ENCODERS

[Stages 1–2 · Edge NPU / MIPI · 30–60 Hz]

Intra-Stage Dataflow: Photodiode integration (t_{exp}) → Zero-copy DMA ring buffers → DINOv2 / ViT patch embeddings → 3D metric unprojection in $SE(3)$.

SPATIAL AFFORDANCES

$$O_t = \langle \mathbf{z}_{feat}, \mathbf{T}_{grasp}, \hat{\mathbf{n}}, \hat{\mu}, r_{safe} \rangle \in SE(3)$$

Instantaneous 60 Hz 3D Affordance Tokens (O_t)

CHAPTER 5: TEMPORAL MEMORY & WORLD MODELS

[Stage 3 State · Host CPU / SRAM · 50 Hz]

Intra-Stage Dataflow: Instantaneous O_t → $SE(3)$ kinematic frame trees ($\mathbf{T}_{tool}^{world}$) → Latent JEPA/RSSM belief dynamics → Uncertainty expansion $\sigma(t) = \sigma_0 + \nu \Delta t$.

WORLD STATE BELIEF

$$S_t = \langle \mathbf{T}_{tool}^{world}, \Sigma_{pos}(t), \mathbf{v}_{target}, \tau_{TTL} \rangle$$

Persistent 50 Hz World Frame Tree & Belief (S_t)

CHAPTER 6: REASONING & INTENT SCHEMAS

[Stage 4 Deliberation · Host MPU · 1–2 Hz (System 2)]

Intra-Stage Dataflow: Human text prompt + World belief S_t → Edge VLM 3D grounding → Kinematic reachability & manipulability → Time-bounded intent lease generation.

EXPIRING INTENT LEASE

$$\mathcal{L}_{intent} = \langle \mathbf{T}_{goal}, \text{BoundingBox}_{3D}, t_{expire} \rangle$$

Time-Bounded 3D Intent Goal Lease ($\mathcal{L}_{intent}$)

CHAPTER 7: TRAJECTORY PLANNING & ACTION CHUNKING

[Stage 5 Trajectory · MPU/NPU · 20 Hz (System 1.5)]

Intra-Stage Dataflow: Intent lease $\mathcal{L}_{intent} + S_t$ → Diffusion Policy / ACT action chunking ($H = 16$) → Quintic splines → Bounded jerk ($\|\ddot{\mathbf{q}}\| \leq j_{max}$).

TRAJECTORY CHUNKS

$$p_t = [\mathbf{q}_1, \dots, \mathbf{q}_H], \|\ddot{\mathbf{q}}\| \leq j_{max}$$

Candidate H -Step Trajectory Proposal Chunk (p_t)

CHAPTER 8: REAL-TIME REFLEX & SAFETY ENFORCEMENT

[Stages 6–7 · Bare-Metal MCU · 1 kHz]

Intra-Stage Dataflow: Candidate p_t → Static SRAM → 1 kHz QP Control Barrier Functions ($h(x) \geq 0$) → Dynamic stopping ($d_{stop} \leq d_{clear}$) → Current clamps.

SAFETY ENFORCEMENT

$$u_t^* = \text{permit}(p_t), \text{WCET} \leq 175 \mu s$$

Permitted Actuator PWM Voltages & Current Clamps (u_t^*)



THE PHYSICAL WORLD (W_{t+1}): IRREVERSIBLE PHYSICAL STATE EVOLUTION

Actuators accelerate mass ($\mathbf{p} = m\mathbf{v}$) · Joule dissipation ($I^2 R t$) · Gear tooth contact forces · Physical universe mutates ($W_t \rightarrow W_{t+1}$)

CLOSED-LOOP CAUSAL FEEDBACK

Moving matter & dissipating heat mutate sensory reality ($W_t \rightarrow W_{t+1} \rightarrow O_{t+1}$)